Politecnico di Milano

SCHOOL OF CIVIL, ENVIRONMENTAL, AND LAND MANAGEMENT ENGINEERING

MASTER OF SCIENCE IN CIVIL ENGINEERING

PROJECT I: MODELLING OF DYNAMIC RESPONSE OF A R.C. STRUCTURE IN THE ELASTIC AND INELASTIC RANGE

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ABSTRACT

The purpose of the report is: 1) to illustrate the criteria adopted for modelling the ELSA frame subject to low dynamic excitation. 2) to illustrate the subsequent modifications of the numerical model set forth in order to improve the correspondence between period of the natural modes from the numerical model and those determined experimentally on the structure. 3) to illustrate the criteria adopted for the modelling of the ELSA frame subject to high excitation (SLU earthquake) and the subsequent modifications eventually made to the numerical model. 4) to carry out the modal analysis with the design response spectrum (EC8) and compare the values obtained for base shear and story displacements with the experimental ones. 5) carry out the dynamic analysis in the time domain with the accelerogram provided and compare the values obtained base shear and story displacements with those in point 4) and with the experimental ones. 6) to set up a nonlinear model for a pushover analysis (modal and constant load distributions) and compare the capacity curve with the experimental results.

CHAPTER 1

INTRODUCTION:

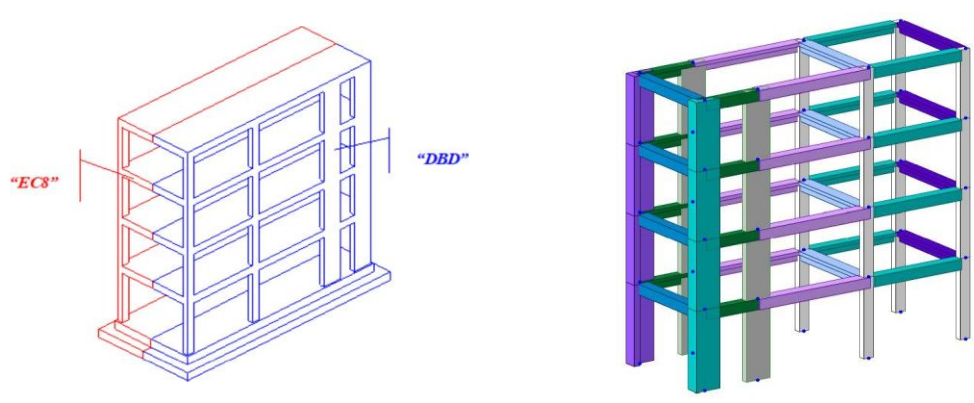
The primary aim of the project is to perform the numerical modelling of an existing reinforced concrete multi-story dual structure (frame-wall). This project intends to illustrate the numerical modelling of an elastic four-storied dual structure (frame-wall) for low and high intensity vibrations. Knowing the results of the experiments we made a step-by-step comparison of the model to achieve the optimal modelling.

Three different models of the structure have been considered:

a. Modal Analysis: Finite Element model with some improvements to attain a value close to the experimental frequencies. The low excitation linear elastic analysis or modal analysis is carried out to approximate the frequencies and the proper modes provided by the experimental data on a model of the structure subjected to the self-weight.
b. Dynamic Analysis: The two different types of seismic loading namely response spectrum and accelerogram, is imposed on the structure and the results of the dynamic analyses are then compared to those provided by experimental pseudo-dynamic tests.
c. Push over analysis: The nonlinear static analysis. Midas Gen has been used to model the structure, different aspects of simulation process, improved geometrical representation of the structure, behavior of the nodes and to the participating mass simulation.

1.1 MODEL GEOMETRY:

A general description of the system is: four-story building, made of two symmetric parts with two design philosophies which are composed by shear walls (rectangular section), L-shaped section and a frame system composed by beams and columns. The general configuration of the building can be seen in the figures given below:



(a) (b)

Figure 1: (a) 3D Model of the given structure (b) 3D Model of structure in MIDAS Gen

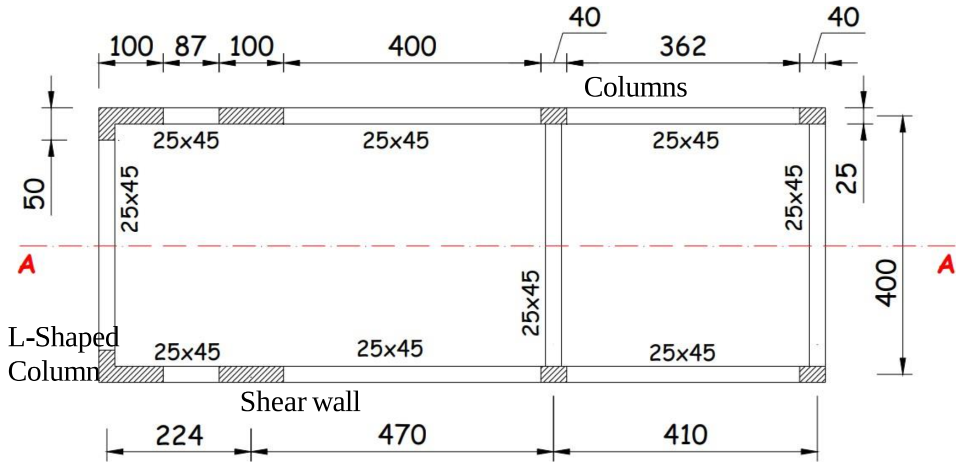


Figure 2: Floor Plan

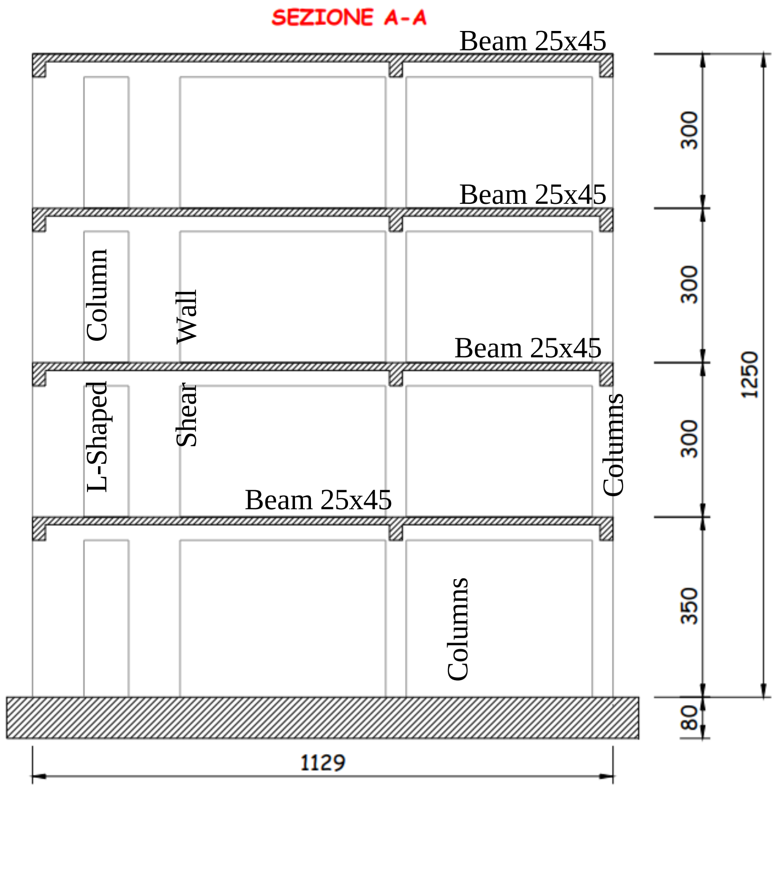
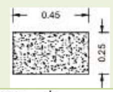
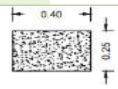
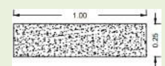
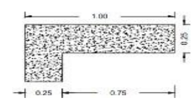


Figure 3: Vertical Section

1.2 Member Dimensions:

The dual structure (frame-wall) system resists the horizontal forces partially by shear walls and partially by moment resisting frames. The geometry of each member is mentioned in table

Table 1: Members Geometry

Member	Model Identification	Geometry
Beams	B 25x45	
Columns	C 25x40	
Shear Wall	C 25x100	
L-shaped Col	L 100x25	

1.3 Material Properties:

1.3.1 Concrete:

The concrete class used in structural design is C20/25 and following are the results obtained from compressive test performed on the cylindrical specimens.

Table 2: Concrete Cylindrical Strength (MPa)

Location	Floor	28-day cubes(average of 2 specimens)	7-month cores (at PsD test)	12-month cores (5 months after PsD test)
Columns	1	22.5	-	-
Columns	2	21.2	-	-
Columns	3	19.7	-	26.5 (1 Core)
Columns	4	24.0	-	-
Slab, Beams	1	22.2	29.3	32.4 (average, 3 cores)
Slab, Beams	2	13.7	31.3	31.2 (average, 3 cores)
Slab, Beams	3	19.8	31.3	34.0 (average, 2 cores)
Slab, Beams	4	18.5	31.9	-
Walls		22.4	-	-
Walls		20.1	-	-
Average		**20.5**	**31.0**	**32.8**

The average value (mean compressive strength) will be taken as 20 MPa henceforth, From mean Compressive strength the characteristic value (lower fractile value) was obtained as,

$$f_{ck} = f_{cm} - 8\ (MPa)$$

$$f_{ck} = 20 - 8$$

$$f_{ck} = 12\ MPa$$

For modulus of elasticity reference can be made to "EN1992-1-1" which gives the following formula for calculation

$$E_{cm}[MPa] = 22000 * \left(\frac{f_{cm}}{10MPa}\right)^{0.3}$$

$$E_{cm}[MPa] = 22000 * \left(\frac{20}{10MPa}\right)^{0.3}$$

$$E_{cm} = 27.085\ GPa$$

1.3.2 Steel:

Tensile test on S500 steel class were performed with different sizes of steel specimens whose results are tabulated as under,

Table 3: Strength and Ultimate Elongation of steel bars

Diameter (mm)	Yield Strength (Mpa)	Tensile Strength (MPa)	Ultimate Elongation A5 (%)
8	557.5	642.1	23.7
10	525.2	617.2	24.2
12	516.5	615	24.8
14	521.9	615.2	26.2
16	506.4	627	24
Average	**525.5**	**623.3**	-

1.4 Dynamic Testing:

A dynamic test considering only the self-weight of the structure with assumption of only linear elastic modelling was carried out. The first 12 Natural frequencies obtained as a result are tabulated as under:

Table 4:Frequencies for Structural Self Weight only

Modes	Frequency (Hz)	Type	Ordering
1	2.06	Direction Y	1
2	3.94	Torsion	1
3	4.31	Direction X	1
4	6.37	Direction Y	2
5	11.62	Direction Y	3
6	12.9	Torsion	2
7	15.32	Direction X	2
8	16.56	Direction Y	4
9	24.31	Torsion	3
10	31.5	Direction X	3
11	34.6	Torsion	4
12	45.87	Direction X	4

The 3 Modes are,

Mode 1: Flexural mode in y-axis

Mode 2: Torsional mode with z-axis as axis of rotation

Mode 3: Translation in x-axis

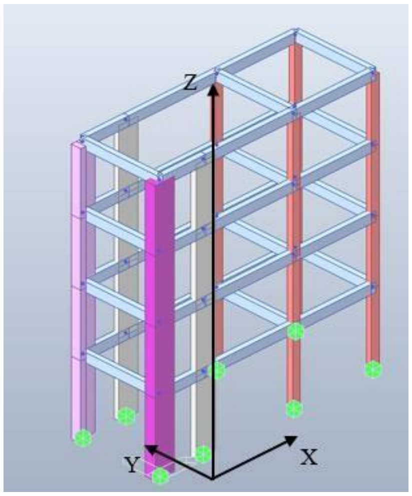


Figure 4:Modes of Vibration

CHAPTER 2

2 MODEL OF STRUCTURE:

A linear elastic model of the structure was made in MIDAS and to better represent the coded natural frequencies were approached by changing various parameters in the following models.

2.1 Model 1:

This first model, the most basic among all, frame elements have been used, applying them the properties of the material C20/25. Because of the roughness of the model, the results were not expected to be precise.

$$m = \frac{\gamma}{g} * Volume \qquad where\ \gamma\ for\ C20/25\ = 20\ KN/m^3$$

$$I = \frac{1}{12} * \mathrm{m} * (\mathrm{L}^2 + \mathrm{W}^2)$$

The data related to the computations is presented in the next tables.

Table 5: (a) Mass and inertia of Slab 1, (b) Slab 2

SLAB 1		
Property	Unit	Value
Length	(cm)	669
Width	(cm)	375
Thickness	(mm)	150
Area	(m^2)	25.09
Volume	(m^3)	3.76
Translation mass	(Ton)	7.67
Rotational Mass	(Ton*m^2)	37.61

(a)

SLAB 2		
Property	Unit	Value
Length	(cm)	385
Width	(cm)	375
Thickness	(mm)	150
Area	(m^2)	14.44
Volume	(m^3)	2.16
Translation mass	(Ton)	4.42
Rotational Mass	(Ton*m^2)	10.63

(b)

The structural model of four-story building, elevation model with the self-weight and top view of slabs made on MIDAS are given as follows

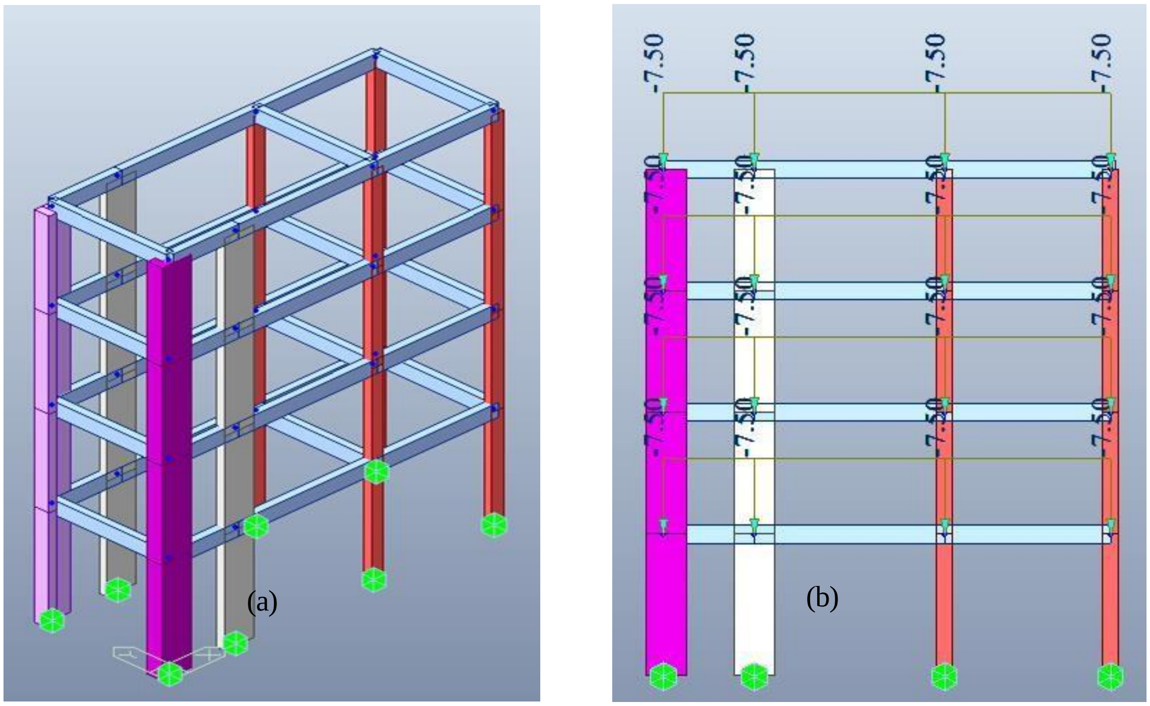


Figure 5:(a) Structural Model (b) Elevation Model

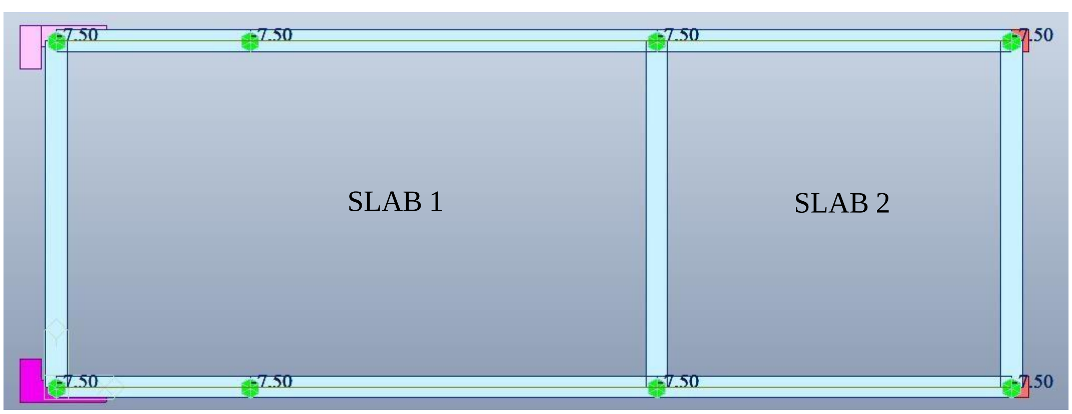


Figure 6: Top view of slab

The following table shows the results of the modal analysis performed, along with the results of the real test, and the average error calculated:

Table 6: Mode Result comparison of model 1

Modes	Type	Exp. Frequency (Hz)	Model Frequency (Hz)	Error (%)
1	Direction Y	2.06	1.7359	15.73
2	Torsion	3.94	2.7148	31.10
3	Direction X	4.31	3.4827	19.19
4	Direction Y	6.37	5.5929	12.20
5	Direction Y	11.62	9.1854	20.95
6	Torsion	12.9	10.3934	19.43
7	Direction X	15.32	12.6474	17.45
8	Direction Y	16.56	15.452	6.69
9	Torsion	24.31	18.297	24.73
10	Direction X	31.5	27.118	13.91
11	Torsion	34.6	28.743	16.93
12	Direction X	45.87	35.797	21.96

The average error calculated of model 1 is given as

$$e = \frac{f_{real} - f_{FEM}}{f_{real}} * 100$$

Table 7: Average Error of Model 1

Avg. Error	
X-direction (%)	18.12777
Y-direction (%)	13.89375
Torsion (%)	23.04747
Total	18.36%

2.2 Model 2:

In this model, to make the structure stiffer, the effect of the slab is considered; to take it into account, the Paulay and Priestley (P&P) formula is followed as:

- Beam surrounded by slab on one side,

$$b_{eff} = \min\left(b_w + \frac{l_x}{24},\ \ b_w + 3h_s,\ \ b_w + \frac{l_{nx}}{4}\right)$$

- Beam surrounded by slab on both sides,

$$b_{eff} = \min\left(\frac{l_x}{8},\ \ b_w + 8h_s,\ \ b_w + \frac{l_{nx}}{2}\right)$$

Where:

- l_x = length of beam
- b_w = width of beam at section's base
- h_s = height of slab
- l_{nx} = distance between beams

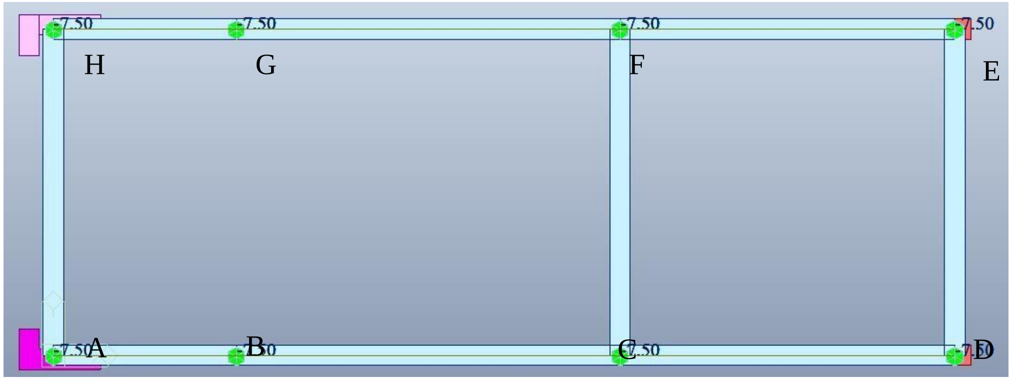


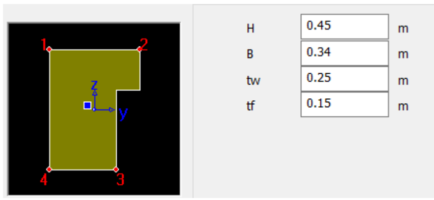
Figure 7: Top view of model

The following table shows the parameters related to the slab and beams:

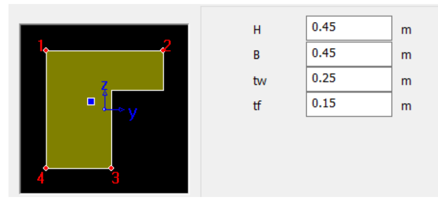
Table 8: Slab and Beam parameters

BEAMS	b_w	l_x	h_s	l_{nx}	b_1	b_2	b_3	b_{eff}
AB & GH	0.25	2.24	0.15	3.75	0.34	0.70	1.19	0.34
BC & FG	0.25	4.70	0.15	3.75	0.45	0.70	1.19	0.45
CD & EF	0.25	4.10	0.15	3.75	0.42	0.70	1.19	0.42
DE	0.25	4.00	0.15	3.85	0.42	0.70	1.21	0.42
CF (T sec)	0.25	4.00	0.15	3.85	0.50	1.45	2.18	0.50
AH	0.25	4.00	0.15	6.69	0.42	0.70	1.92	0.42

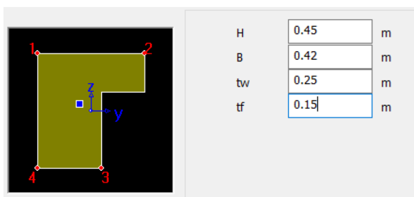
AB & GH



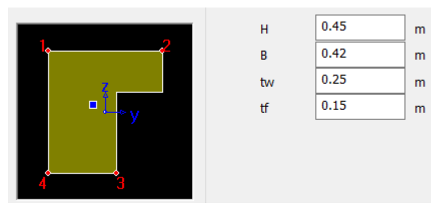
BC & FG



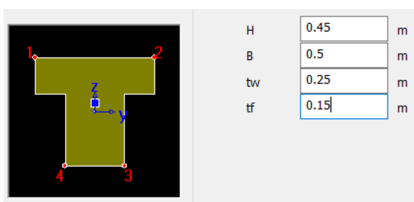
CD & EF



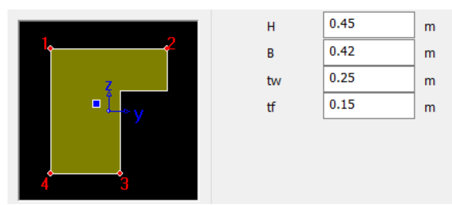
DE

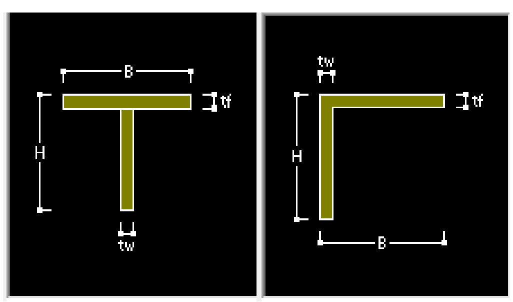


CF



AH





LEGEND

Table 9: Properties of (a) Slab 1, (b) Slab 2

SLAB 1		
Property	Unit	Value
Length	(cm)	639.50
Width	(cm)	357.00
Thickness	(mm)	150.00
Area	(m^2)	22.83
Volume	(m^3)	3.42
Translation mass	(Ton)	6.98
Rotational Mass	(Ton*m^2)	31.21

(a)

SLAB 2		
Property	Unit	Value
Length	(cm)	355.50
Width	(cm)	341.00
Thickness	(mm)	150.00
Area	(m^2)	12.12
Volume	(m^3)	1.82
Translation mass	(Ton)	3.71
Rotational Mass	(Ton*m^2)	7.50

(b)

The results obtained with the new model are given in the following table:

Table 10: Mode Result comparison of model 2

Modes	Type	Exp. Frequency (Hz)	Model Frequency (Hz)	Error (%)
1	Direction Y	2.06	1.7553	14.79
2	Torsion	3.94	2.7808	29.42
3	Direction X	4.31	3.5243	18.23
4	Direction Y	6.37	5.5881	12.27
5	Direction Y	11.62	9.2967	19.99
6	Torsion	12.9	10.2258	20.73
7	Direction X	15.32	12.6012	17.75
8	Direction Y	16.56	15.0059	9.38
9	Torsion	24.31	18.1886	25.18
10	Direction X	31.5	26.6282	15.47
11	Torsion	34.6	28.2105	18.47
12	Direction X	45.87	32.939	28.19

The average error calculated by the model 2 is given as:

Table 11: Average Error of second model

Avg. Error	
X-direction (%)	19.90825
Y-direction (%)	14.11116
Torsion (%)	23.44972
Total Error	19.15%

2.3 Model 3:

2.3.1 Model 3(a): Beam end offset/rigidity factors:

This model has been modified from the 3.2. Initial model 2 (P&P), by modifying the stiffness of the node. From this model, offsets have been inserted in Midas-Gen considering the real width of the members, as illustrated below:

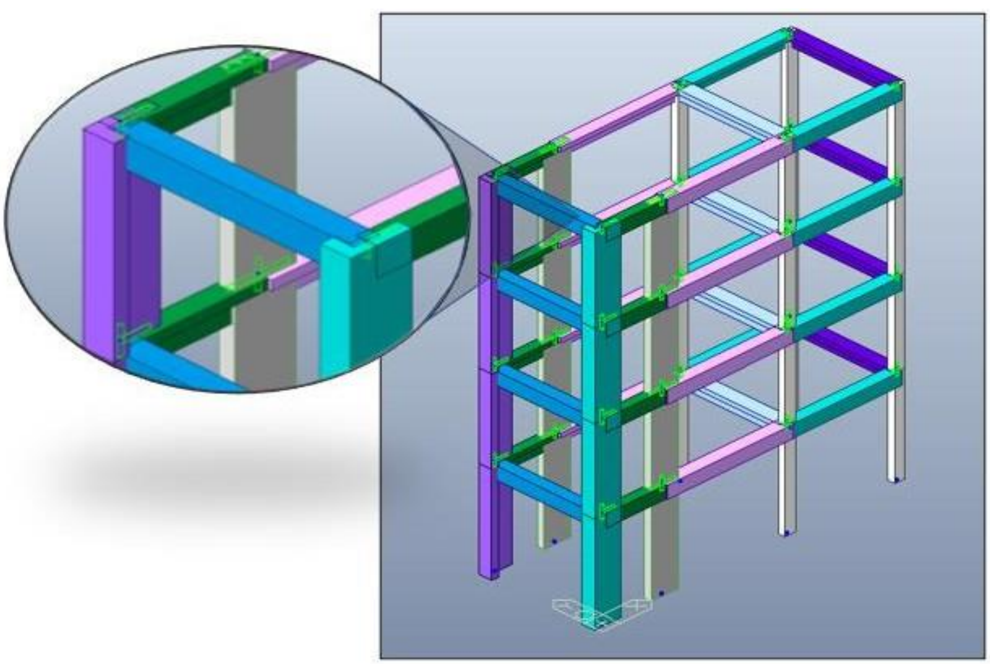


Figure 8: Model with Beam End offsets

Considering the improvements our computations are as follows:

Table 12: Rigidity factors of members

Members	RGD I (Near end)	RGD j (Far end)
Columns (Ground floor)	0	0.225
Columns (other floors)	0.225	0.225
Longitudinal Beams span 1	0.5	0.5
Longitudinal Beams span 2	0.5	0.2
Longitudinal Beams span 3	0.2	0.2

The model 3(a) under modal analysis gives the following results:

Table 13: Model 3(a) frequencies compared with experimental data

Modes	Type	Exp. Frequency (Hz)	Model Frequency (Hz)	Error (%)
1	Direction Y	2.06	1.9096	7.30
2	Torsion	3.94	3.204	18.68
3	Direction X	4.31	4.6625	-8.18
4	Direction Y	6.37	6.2452	1.96
5	Direction Y	11.62	10.6891	8.01
6	Torsion	12.9	12.0251	6.78
7	Direction X	15.32	16.134	-5.31
8	Direction Y	16.56	18.3052	-10.54
9	Torsion	24.31	21.4222	11.88
10	Direction X	31.5	33.2539	-5.57
11	Torsion	34.6	33.7287	2.52
12	Direction X	45.87	34.034	25.80

The average error calculated by the model 3(a) is given as:

Table 14: Average error of the frequency

Avg. Error	
X-direction (%)	1.685863
Y-direction (%)	1.683174
Torsion (%)	9.964911
TOTAL	4.444649

2.3.2 Model 3(b): Panel offset

This improved model has been developed considering that in real case the zone around the nodes is not perfectly rigid, a panel offset factor regarding the stiffness of that linking can be set, being this value different from the unity. From this model, panel offsets have been inserted in Midas-Gen, as illustrated below:

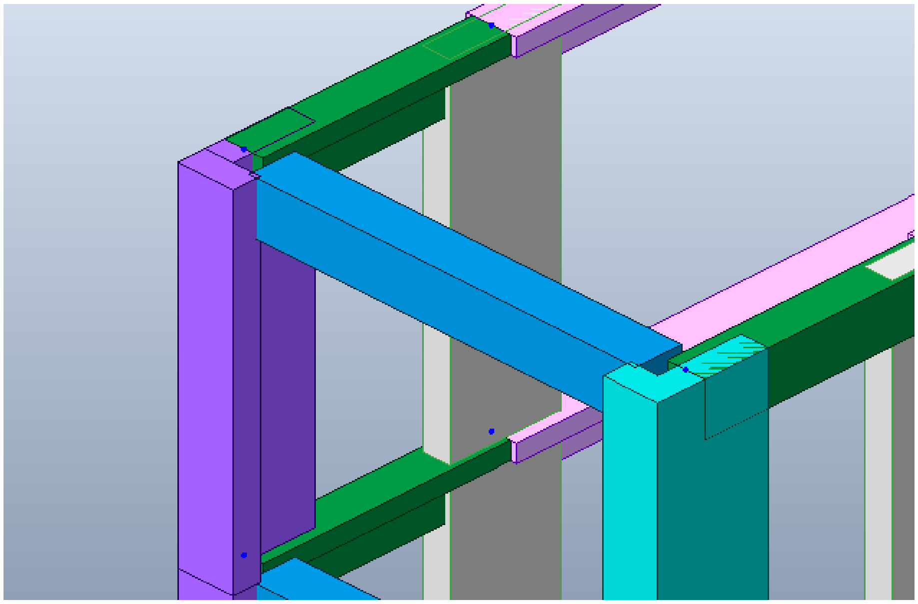


Figure 9: Panel offset

Considering the improvements our computations are as follows:

Table 15: Natural frequencies vs Panel offset factors

	PANEL OFFSET FACTORS				
	0	0.5	0.7	0.8	1
NATURAL FREQUENCIES	1.9096	1.9824	2.0044	2.0154	2.0374
	3.204	3.3173	3.3573	3.3782	3.4217
	4.6625	4.6841	4.6914	4.6951	4.7027
	6.2452	6.4353	6.4882	6.5148	6.5681
	10.6891	11.0212	11.1309	11.1879	11.3061
	12.0251	12.2693	12.3227	12.3499	12.4051
	16.134	16.2317	16.2465	16.254	16.2692
	18.3052	18.5625	18.5909	18.6052	18.634
	21.4222	21.8494	21.9628	22.0221	22.1459
	33.2539	33.5306	33.5466	33.5547	33.571
	33.7287	34.3247	34.3546	34.3696	34.3999
	34.034	34.4528	34.5165	34.5496	34.618

The model 3(b) under modal analysis gives the following results:

Table 16: Model 3(b) frequencies compared with experimental data

Modes	Type	Exp. Frequency (Hz)	Model Frequency (Hz)	Error (%)
1	Direction Y	2.06	2.0044	2.70
2	Torsion	3.94	3.3573	14.79
3	Direction X	4.31	4.6914	-8.85
4	Direction Y	6.37	6.4882	-1.86
5	Direction Y	11.62	11.1309	4.21
6	Torsion	12.9	12.3227	4.48
7	Direction X	15.32	16.2465	-6.05
8	Direction Y	16.56	18.5909	-12.26
9	Torsion	24.31	21.9628	9.66
10	Direction X	31.5	33.5466	-6.50
11	Torsion	34.6	34.3546	0.71
12	Direction X	45.87	34.5165	24.75

The average error calculated by the model 3(a) is given as*:*

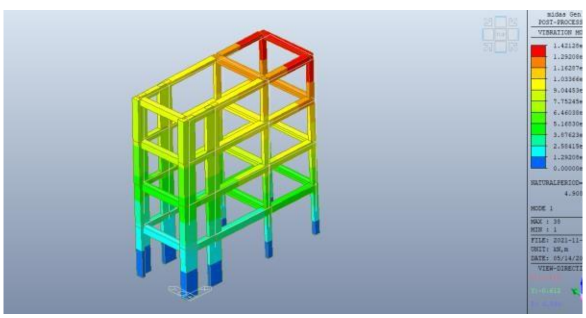
Table 17: Average Error of model 3(b)

Avg. Error	
X-direction (%)	0.839373
Y-direction (%)	-1.80283
Torsion (%)	7.407267
Total	2.147937

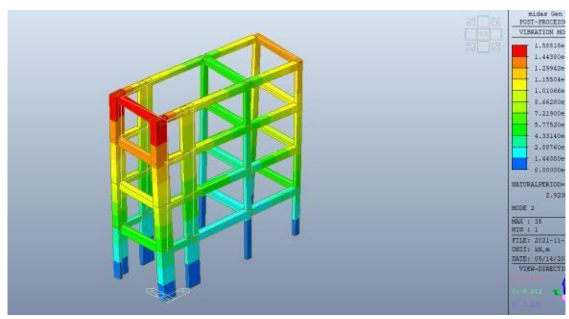
The obtained mean error, around 2.15% is acceptable for our purposes, so this model can be considered definitive for the next analyses.

2.4 Mode shapes:

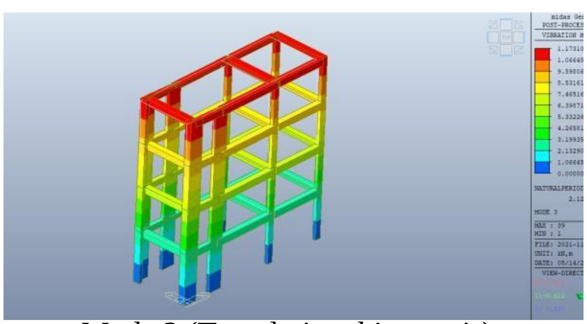
Mode shapes from MIDAS Gen, for the 12 modes along with legends are given below.



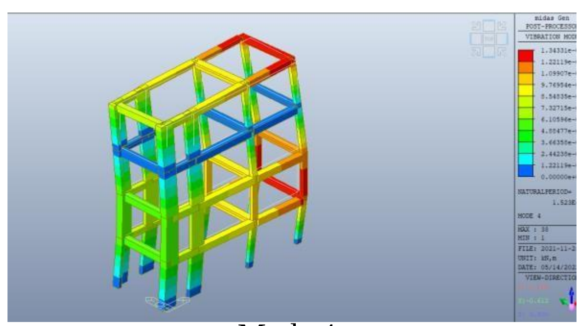
Mode 1 (Translational y-axis)



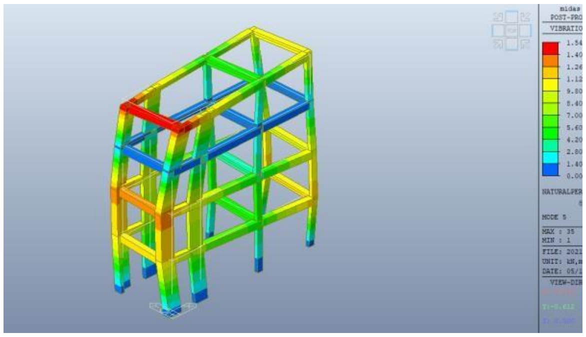
Mode 2 (Torsional in z-axis)



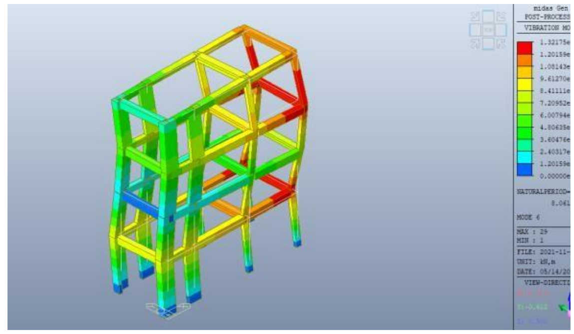
Mode 3 (Translational in x-axis)



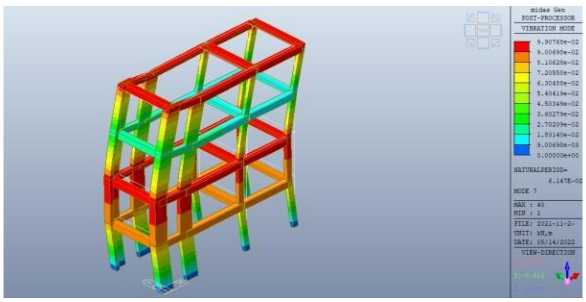
Mode 4



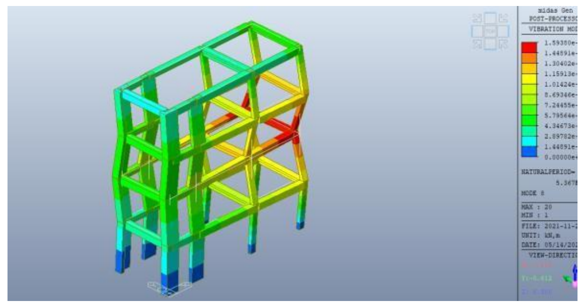
Mode 5



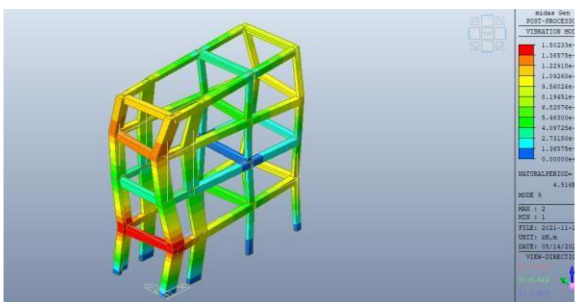
Mode 6



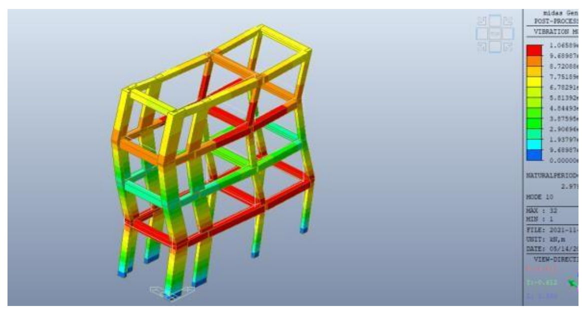
Mode 7



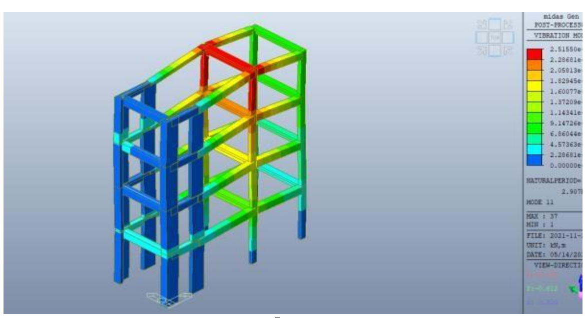
Mode 8



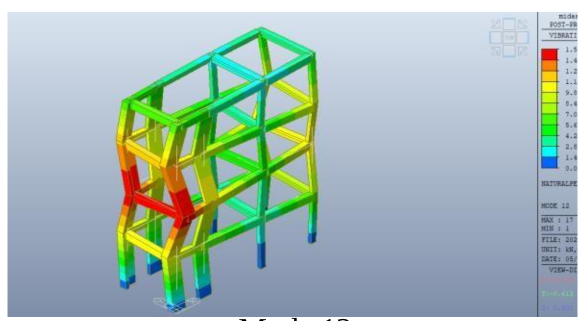
Mode 9



Mode 10



Mode 11



Mode 12

Figure 10: Mode Shapes

CHAPTER 3

3.1 ELASTIC MODEL FOR DYNAMIC TESTING AT ULS:

The experimental results here forth presented were obtained through a pseudo dynamic test on a real structure.

The experimental model was subjected to dynamic excitation pulses, to simulate the response of the modelled structure, with respect to the Ultimate Limit State (ULS), applying the given seismic action in the form of two types of input:

a. Response spectrum
b. Accelerogram.

The real structure was experimented with the pseudo-dynamic test in which the model was analyzed in non-linear field after the formation of plastic hinges at the critical sections. In Pseudo dynamic test varying loads are applied to the structure and the motion and deformation are used to model the behavior of the structure under the action of a seismic event.

During the experiment two quantities were measured using a set of transducers and actuators:

a. Absolute Displacements.
b. Applied Force.
c.

3.1.1. Considerations For The Experimental Test:

Five tests were carried out, characterized by the following parameters:

- Duration of the earthquake.
- Max ground acceleration induced by the earthquake (PGA)
- PGA amplification factor (Scaling factor)

Table 18: Pseudo dynamics tests

Test	Type	Scaling Factor for Design Earthquake	PGA	Duration
D01		Modal Data		
D02	Pseudo dynamic	0.05	0.028 g	8 sec
D03	Pseudo dynamic	0.10	0.056 g	15 sec
D04	Pseudo dynamic	0.15	0.085 g	4 sec
D05	Pseudo dynamic	1.00	0.56 g	15 sec
D06	Pseudo dynamic	1.50	0.85 g	6 sec

The test D05 corresponds to the local collapse and has a PGA equal to 0.56 g, has a response spectrum having a plateau of 1 g, equal to that of the EC8 design spectrum.

For the analysis of the structure two methods were followed:

- Analysis with accelerogram (Time history)
- Analysis with the response spectrum

3.1.2. Time History and Response Spectrum Analysis:

The experimental results obtained through the Pseudo Dynamic Test (PsD), the dynamic response of a certain structure without the necessity of installing shaking tables, by means of the resolution of the equations of motion into certain time steps using time integration algorithm. For this test, the structure is subjected to the next accelerogram:

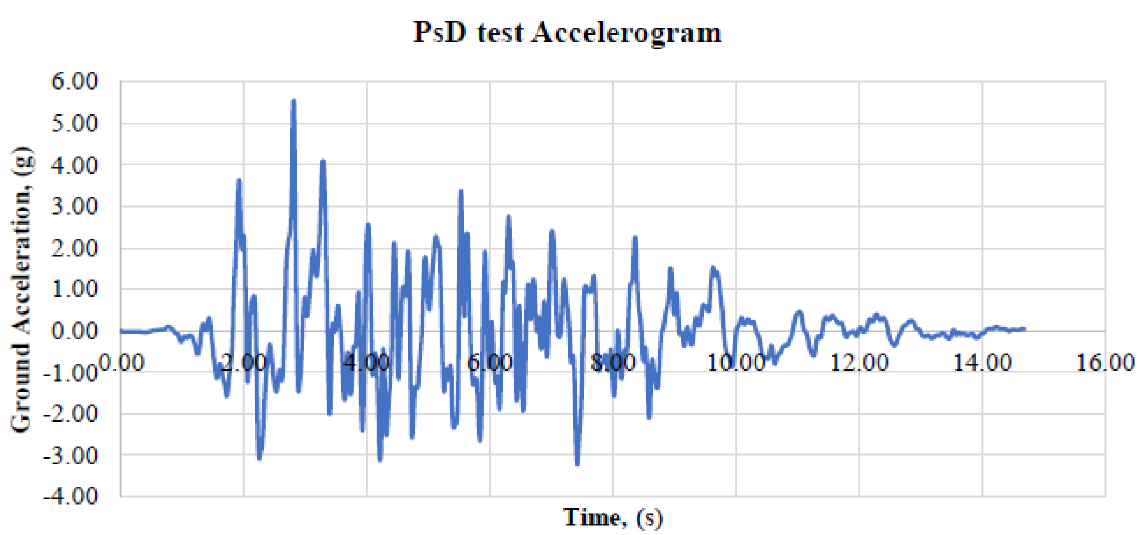


Figure 11: PsD test Accelerogram

In this other graph, the response spectrum for a damping of 5% is presented. The Response spectrum is defined as the curve described by the maximum absolute value of the relative displacement v(t) as the period varies.

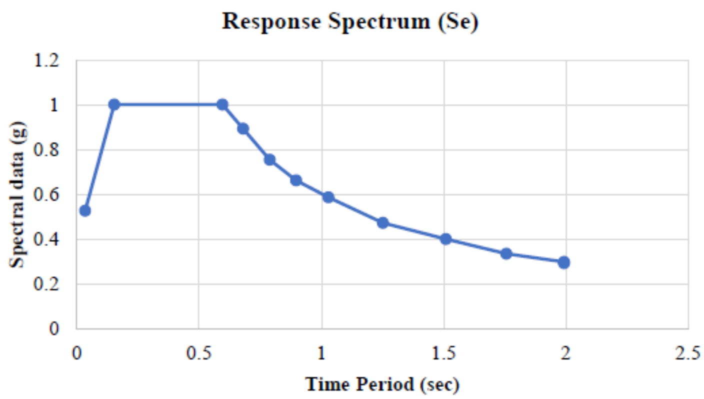


Figure 12: Response Spectrum

3.2. Modification to Plastic Behavior of Structure:

The earthquake load occurred in the x-direction of the frame while the transverse displacements were constrained. Internal forces/reactions were measured by the actuators.

Because of practical limitation of the structure the differences in floor masses simulated in computer during test and the frame was applied only in part by water containers on each floor. This created a difference in axial load which could affect the non-linear model. Therefore, a mass distribution is presented as follows to represent the structural and added weights to perform test.

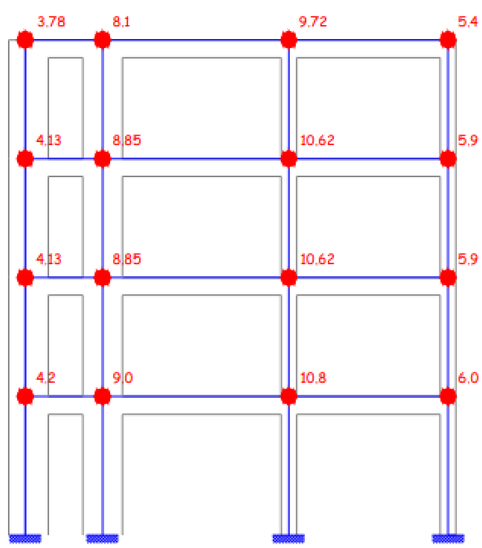


Figure 13-Additional nodal masses

The previously developed models are in linear elastic regime; cracking is not considered, also the moment of inertia of full section is considered. This hypothesis is only acceptable for serviceability or small displacements, but for ultimate analysis (such as in our case, the structure under seismic actions) strong cracking and damaging of the joints may arise.

In the “new” model the two masses positioned on each floor have been replaced with the mass distribution shown in below figure, which represents the structural weight and the added weight to perform the test. Furthermore, for each mass, the inertia has been computed with respect to the floor's center of gravity.

3.3. Ductility and Cracking:

The seismic building design must bring ductility into account. Given the cracking, certain extra energy dissipation resources connected to it gives yield reinforcement. The evaluation of the seismic reaction in relation to the ultimate limit state for life safety considers the origins of structural ductility, and as a result, the analysis would not be linear, but a linear analysis based on the ideas that. The behavior factor q may also be introduced. The behavioral factor q stands for the percentage of the values of the seismic forces to which a correction factor is the resistance of the structure would be entirely elastic, and the clearly reflected of the earthquake forces. To take this into account, the inertia of the beam elements should be reduced up to the cracked one, and the rigidity of the nodes should also be reduced. This has been carried out in the following models.

To clearly understand this concept lets understand F-u behavior of two simple oscillators (1 DOF systems) with same geometry, mass but different constitutive laws subjected to a monotone push until the maximum displacement is reached. One is characterized by a linear elastic behaviour, while the other by an elastic perfectly plastic behaviour.

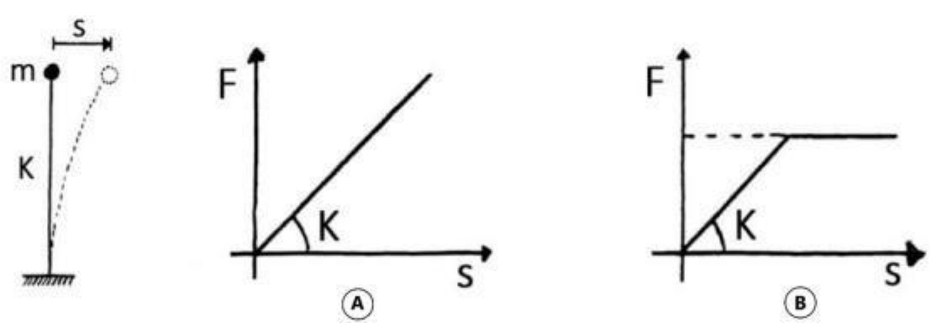


Figure 14- Linear Elastic Behavior (A) Elasto-plastic behavior (B)

If $T0$ > 0.5s : the systems get the same value for the maximum displacement and the yielding limit is located at $upl{-}max/\mu d$. Referring to Figure below, it is possible to see that the ratio between the two ultimate force values represent exactly the behaviour factor q previously defined. In this case Equal displacement rule holds:

$$q =$$

$$u_{pl-max} = u_{el-max} \cdot \frac{\mu d}{q} = u_{el-max}$$

$$F_{el-max} = F_{pl-max} \cdot q$$

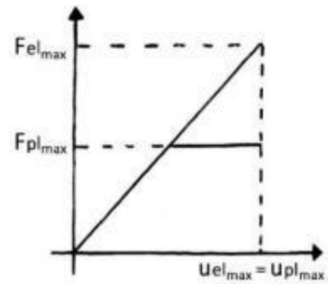


Figure 15- Response To>0.5

If 0.1 s < T <0.5 the systems get the same energy (the area below the two paths is the same). According to the equal energy principle (of Newmark), imposing Tc= 0.5 s, in this case we have:

$$F_{el-max} = F_{pl-max} \cdot q$$

$$u_{pl-max} = u_{el-max} \cdot \frac{\mu d}{q} = u_{el-max}$$

$$\mu d = 1 + (q - 1) \cdot \frac{TC}{T0}$$

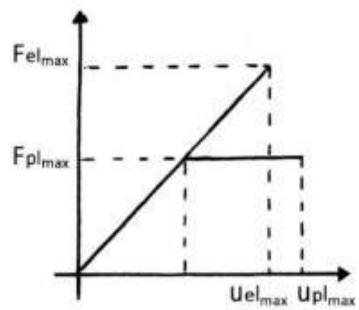


Figure 16- Response 0.1<To<0

If $T0$ < 0.1s : given the small natural period, 2 systems have the same maximum force. In this case (Figure 3-6) there is no benefit in ductility and the entire resistance is provided by the elastic field. Maximum force conservation holds:

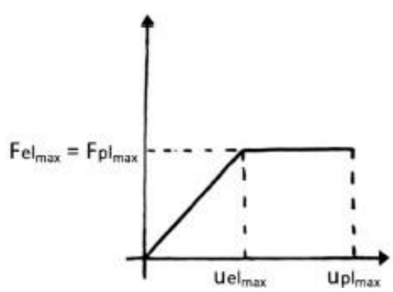


Figure 17- Response To<0.1

Thus, Shear at the base and displacement at the top are the main studies. Depending on the first critical mode of the structure T0 in the considered direction, following expressions will be applied:

$$\mu_D = q \qquad \text{per} \quad T_0 > T_C \qquad \text{"Equal displacement rule"}$$

$$\mu_D = 1 + (q-1)\frac{T_C}{T_0} \qquad \text{per } T_0 < T_C \quad \text{"Equal energy rule"}$$

Value of Tc will be assumed as per EC8 to be 0.5 s, which is chosen according to the type of soil, for our case the soil type is B. But for the response spectrum of a seismic event and with careful analysis it is decided, as a period coinciding with the end of plateau, Tc = 0.6 s.

On the other hand, T0 varies according to the changes made for each model, for cracking, by decreasing the rigidity of the construction elements. The reduction in rigidity will be different for each model discussed below, introducing different percentages of reduction for each type of structural element. The NTC2008 and UNI EN 1998 do not give any precise indication for the adaptation of stiffness value, they suggest a 50% reduction in the rigidity for cracked elements. Following some indications in the FEMA 356 (Prestandard and Commentary for the Seismic Rehabilitation of Buildings) the following table is used.

Chapter 6: Concrete

Table 6-5 Effective Stiffness Values

Component	Flexural Rigidity	Shear Rigidity	Axial Rigidity
Beams—nonprestressed	$0.5E_cI_g$	$0.4E_cA_w$	—
Beams—prestressed	E_cI_g	$0.4E_cA_w$	—
Columns with compression due to design gravity loads $\geq 0.5\ A_gf'_c$	$0.7E_cI_g$	$0.4E_cA_w$	E_cA_g
Columns with compression due to design gravity loads $\leq 0.3\ A_gf'_c$ or with tension	$0.5E_cI_g$	$0.4E_cA_w$	E_sA_s
Walls—uncracked (on inspection)	$0.8E_cI_g$	$0.4E_cA_w$	E_cA_g
Walls—cracked	$0.5E_cI_g$	$0.4E_cA_w$	E_cA_g
Flat Slabs—nonprestressed	See Section 6.5.4.2	$0.4E_cA_g$	—
Flat Slabs—prestressed	See Section 6.5.4.2	$0.4E_cA_g$	—

Note: It shall be permitted to take I_g for T-beams as twice the value of I_g of the web alone. Otherwise, I_g shall be based on the effective width as defined in Section 6.4.1.3. For columns with axial compression falling between the limits provided, linear interpolation shall be permitted. Alternatively, the more conservative effective stiffnesses shall be used.

Figure 18: Effective Stiffness values (FEMA 356)

3.4. Analysis:

The application of time history and spectrum to the various numerical model having different flexural and shear rigidity are discussed here and a comparison between numerical and experimental data is made.

3.4.1. Existing Model 1 (Ref Model 3.2):

First, the existing model obtained in the previous sections is analyzed without changing stiffness of elements. Updating the masses and ultimately the rotational inertia as per considerations in the table.

Structural period is modified and the first mode in x-direction results to have a period of T0 = 0.351s which is less than Tc = 0.6 s so we can apply the equal energy rule.

Parameter of tables:

Table 19- Model Parameter

q	5
Tc	0.6
T_0	0.351
μ_d	7.837607

Table 20- Result Comparison of model 1

Analysis Type	F (el-max)	F (pl-max)	Error (%)	μ (el-max)	μ(pl-max)	Error (%)
RS	2573	514.6	55.71047	40.7	63.79812	-36.6384
Time History	2882	576.4	50.3916	41.9	65.67915	-34.7703
Experimental		1161.9			100.689	

Figure 19: Model 1 (Top Displacement)

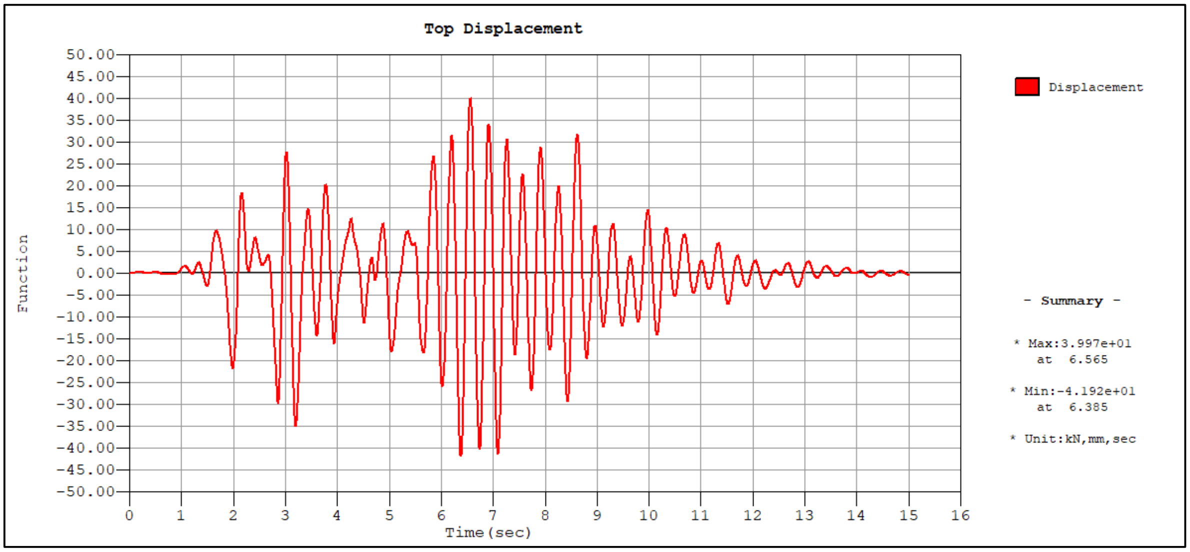
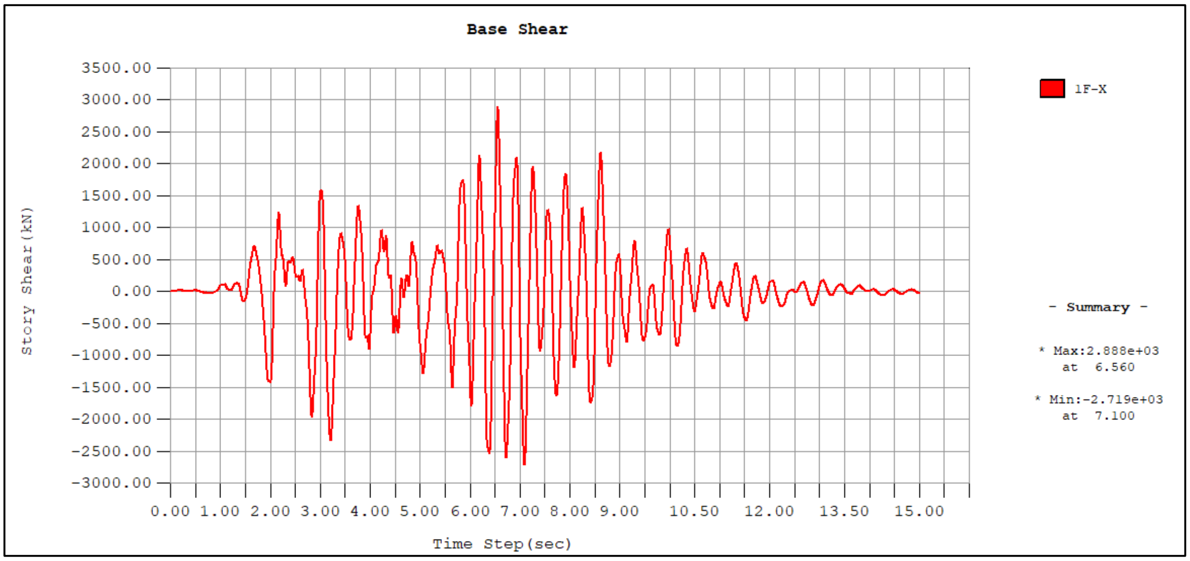


Figure 20: Model 1 (Base Shear)



3.4.2. Model 2:

In the subsequent models flexural and shear rigidities were reduced subsequently in accordance with the indications in the FEMA 356 (Pre-standard and Commentary for the Seismic Rehabilitation of Buildings).

Structural period is modified and the first mode in x-direction has a period of T0 = 0.4234s which is less than Tc = 0.6 s so we can again apply the equal energy rule.

Table 21- Model Rigidity parameter

Members	Bending (EI)	Shear (GA)
BEAMS	0.5	0.4
COLUMNS	0.5	0.4
WALL	1	1

Parameter of tables:

q	5
Tc	0.6
T_0	0.4234
μ_d	6.668399

Table 22- Result comparison of model 2

Analysis Type	F (el-max)	F (pl-max)	Error (%)	μ (el-max)	μ (pl-max)	Error (%)
RS	3070	614	47.16	71.7	95.625	-5.030
Time History	2884	576.8	50.36	60.3	80.421	-20.129
Experimental		1161.9			100.689	

The error percentage for shear at the base does not change significantly while the displacement error percentage has decreased for both the cases.

Figure 21: Model 2 (Top Displacement)

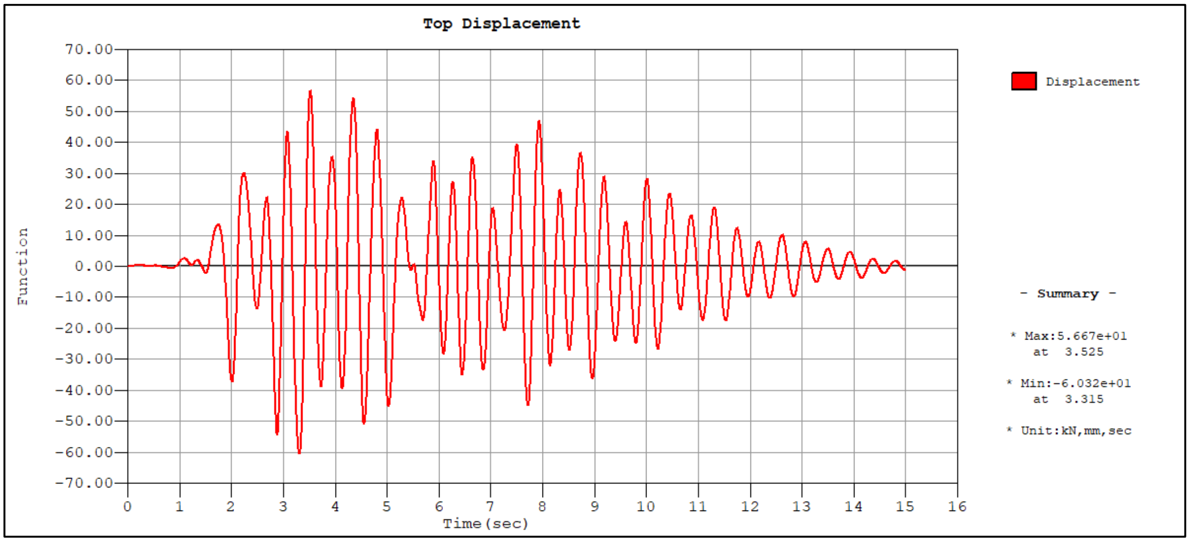
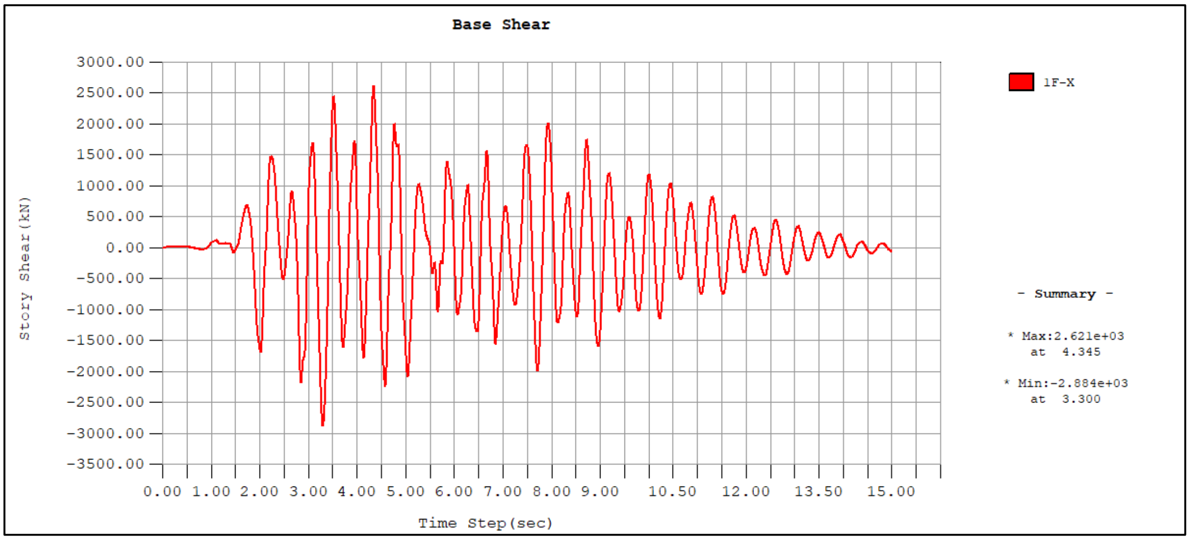


Figure 22: Model 2 (Base Shear)



3.4.3. Model 3:

In this model the rigidities of shear walls were reduced to recommended ones because of their impact on the bearing of actions.

Structural period is modified and the first mode in x-direction has a period of T0 = 0.4468s which is less than Tc = 0.6 s so we can again apply the equal energy rule.

Members	Bending (EI)	Shear (GA)
Beams	0.5	0.4
Columns	0.5	0.4
Wall	0.5	0.8

Parameter of model:

q	5
Tc	0.6
T_0	0.4468
μ_d	6.371530886

Table 23- Result Comparison of model 3

Analysis Type	F(el- max) (KN)	F(Pl-max) (KN)	Error (%)	μ(el- max) (mm)	μ(pl-max) (mm)	Error (%)
RS	3120.1	624.02	46.29314	79.3	101.05248	0.360993
Time History	2750	550	52.66374	65.8	83.8493465	-16.7244
Experimental		1161.9			100.689	

The shear at the base does not undergo significant variations, while the displacement tends to approach the experimental values in both cases, remaining however still distant.

Figure 23: Model 4 (Top Displacement)

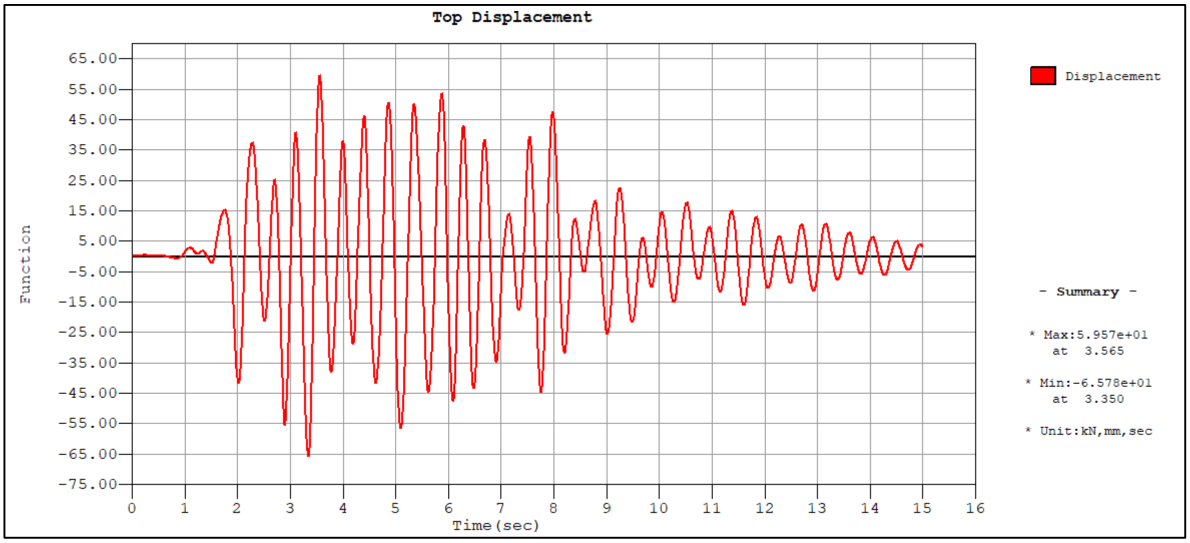
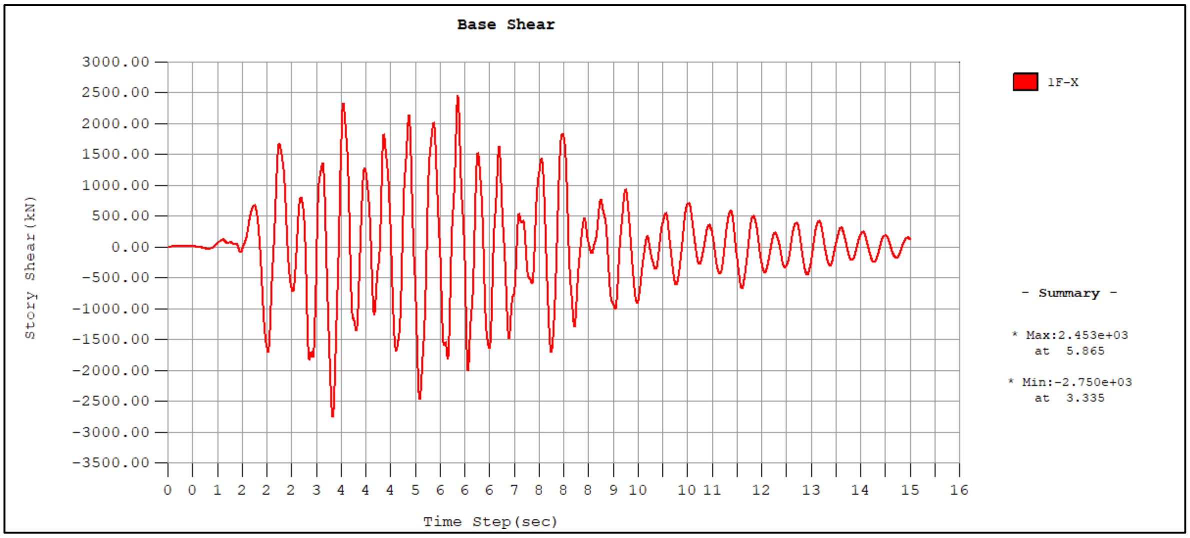


Figure 24: Model 4 (Base Shear)



3.4.4. Model 4:

The following further changes were made to the rigidities.

Table 24- Rigidity of Elements

Members	Bending (EI)	Shear (GA)
Beams	0.5	0.4
Columns	0.5	0.4
Wall	0.5	0.4

The first period is T0 = 0.5060 s which is less than TC = 0.6s and thus we will still use equal energy rule.

Model Parameters

q	5
Tc	**0.6**
T0	**0.506**
μd	**5.743083**

Table 25- Comparison of model 4

Analysis Type	F(el- max) (KN)	F(Pl-max) (KN)	Error (%)	μ(el-max) (mm)	μ (pl- max) (mm)	Error (%)
RS	2988	597.6	48.567002	90.6	104.064664	3.352565
Time History	2486	497.2	57.208021	86.4	99.24047431	-1.43861
Experimental		1161.9			100.689	

We have reached the experimental value of the displacement and error percentage is 3% maximum but for the shear error at the base, we do not see any significant changes.

Figure 25: Model 4 (Top Displacement)

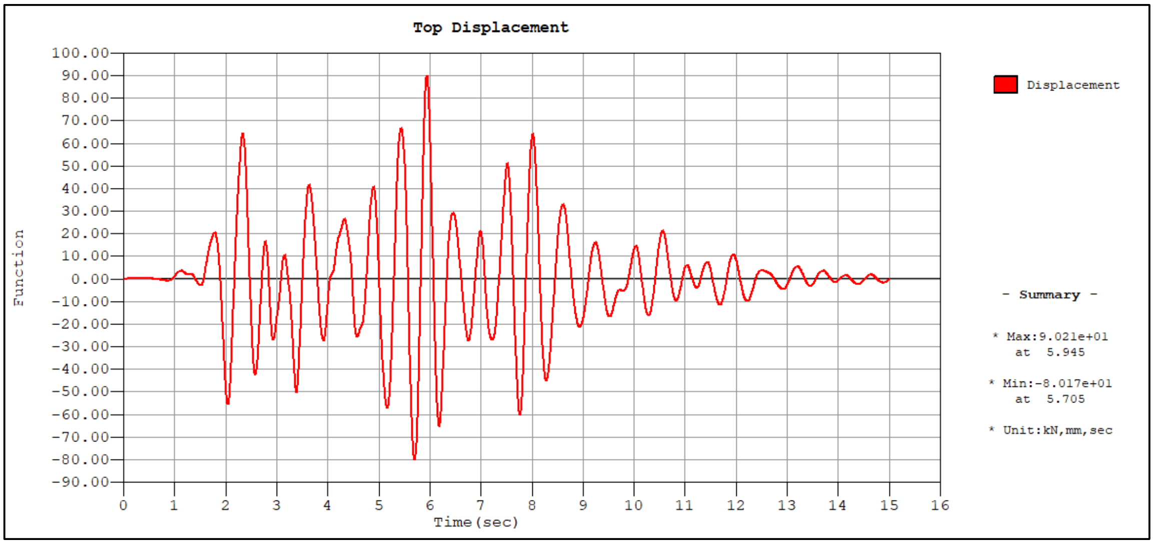
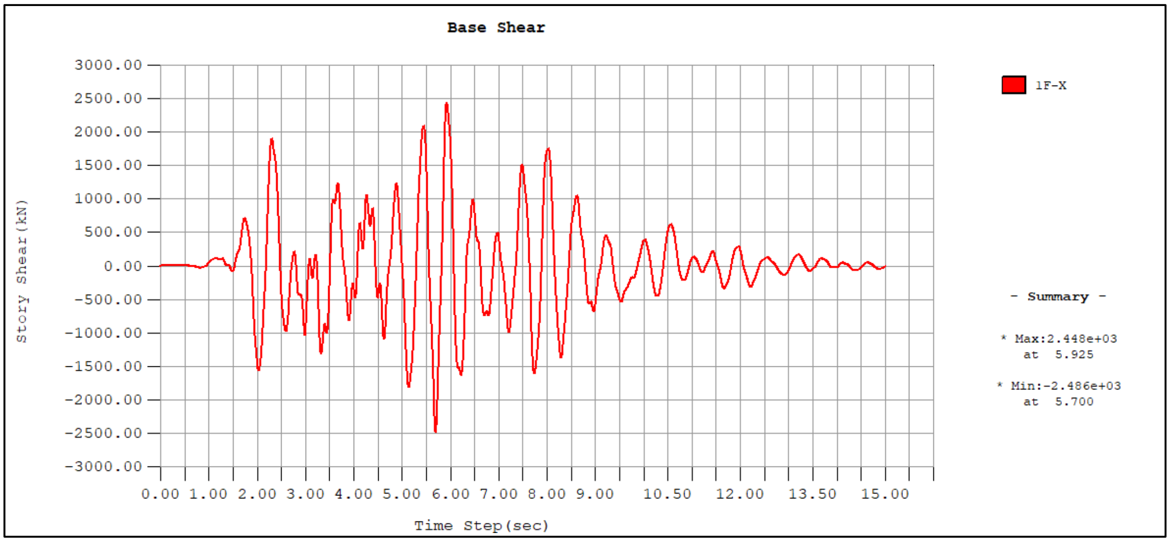


Figure 26:Model 4 (Base Shear)



Modified Behavior Factor q:

Finally, errors of displacement values were decreased in both analyses, while results of the base shear still differ significantly from the experimental value. The reason could be that the structure does not show the initially hypothesized ductility and consequently the behavior factor is incorrectly defined. Approximate value of the behavior factor could be computed as a ratio between the average base shear $Fel{-}max$ from the two linear analyses and the experimentally measured one $Fexp$.

$$q = \frac{F_{el-max}}{F_{experiment}}$$

$$q = \frac{2737}{1161.9} = 2.35$$

$$\mu_d' = 1 + (2.35 - 1) * \frac{0.6}{0.506}$$

$$\mu_d' = 2.6$$

Applying these factors to the obtained parameters from model, new maximum values of base shear and displacement are in table below.

Table 26- Results Comparison of final model with q=2.35

Analysis Type	F(el-max) (KN)	F(Pl-max) (KN)	Error (%)	µ(el- max) (mm)	µ(pl- max) (mm)	Error (%)
RS	2836.45	1207	-3.88157	83	94	-6.6432281
Time History	2780.05	1183	-1.81599	87	97.7	-2.9685467
Experimental		1161.9			100.689	

The most suitable model for the high intensity test is model 2.2, which captures quite good the initial response of the structure during the dynamic event and the non-linear behavior progressively. Elastic results from the model are analyzed and transformed in plastic ones via behavior and ductility factors.

CHAPTER 4

PUSHOVER ANALYSIS:

It is a non-linear static analysis whereby structure is subjected to horizontal static forces stemming from the earthquake's dynamic action. This helps in capturing the plasticization and high displacements behavior of structure due to ductility of elements.

The Pushover process comprises of increasing forces under specified distribution meanwhile monitoring two parameters.

i) Shear at Base
ii) Displacement of top (Max displacement of structure)

A curve set between these two parameters is called a capacity curve. The pushover curve gives an indication of the failure modes of structure by tracking the hinge formation in the critical joints.

Following steps were included in Pushover analysis.

4.1 Pushover Global Control:

Consisted of determining the initial control conditions/operations including,

- Dead and Self weight were included in the load cases.
- For convergence Failure, Max iterations and sub steps were fixed to 10.

4.2 Pushover Load Cases:

As per code requirements, two load distributions were considered.

- Constant (Uniform acc.) load pattern.
- Modal load pattern (considering mode 3, X-direction).

As building is non symmetric the positive and negative responses will differ hence combining with the above two patterns a total of four capacity curves are expected.

The considered load patterns are as follows,

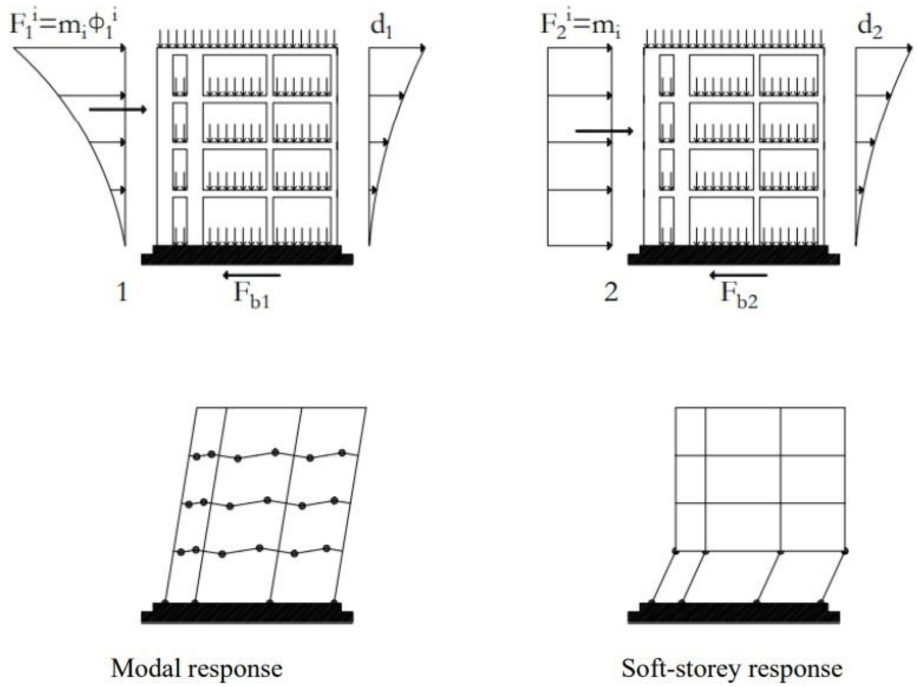


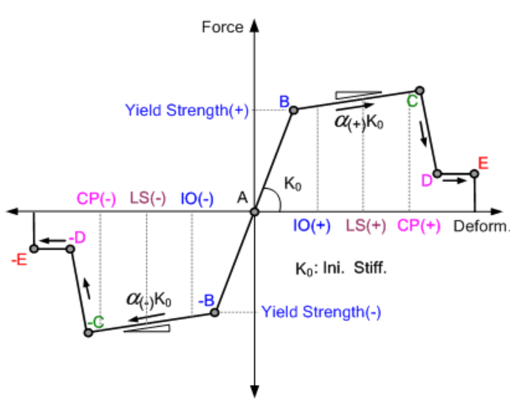
Figure 27: Modal and Uniform load distribution patterns

4.3 Pushover Hinge Properties:

Plastic properties are considered to be lumped at member ends and as per EC8 prescriptions were divided into following categories.

- Beams
- Columns
- Shear walls

Figure 28:Assigning Hinge Properties (FEMA356)



Structural Reinforcement were assigned as per provided specification of EC-8 and DBD as the plastic hinges considers the x-section properties.

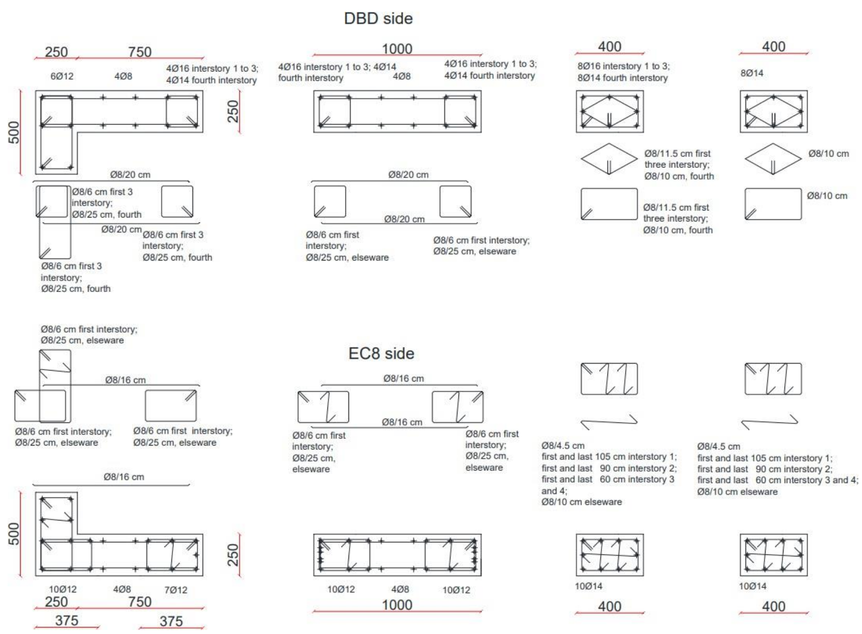
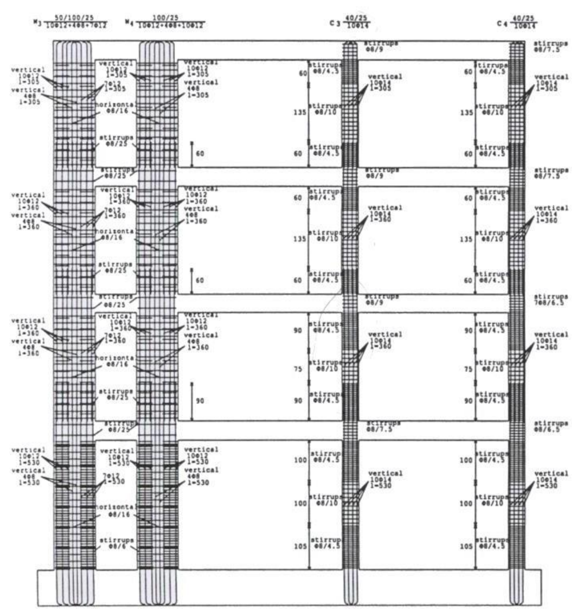


Figure 29: As per Colombo et al. (2002) and Martinelli (2004)



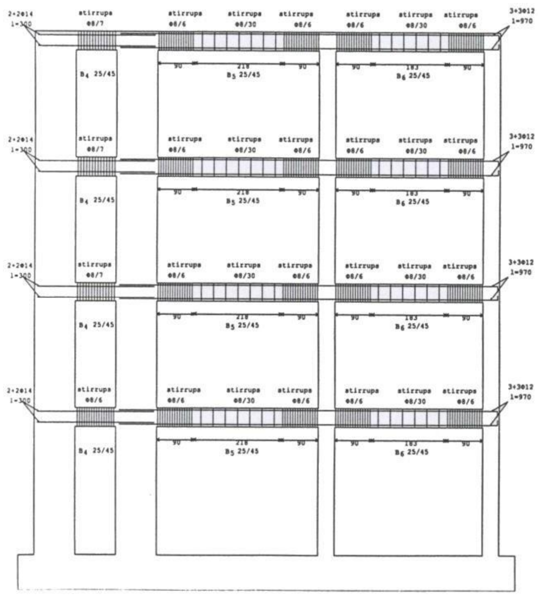
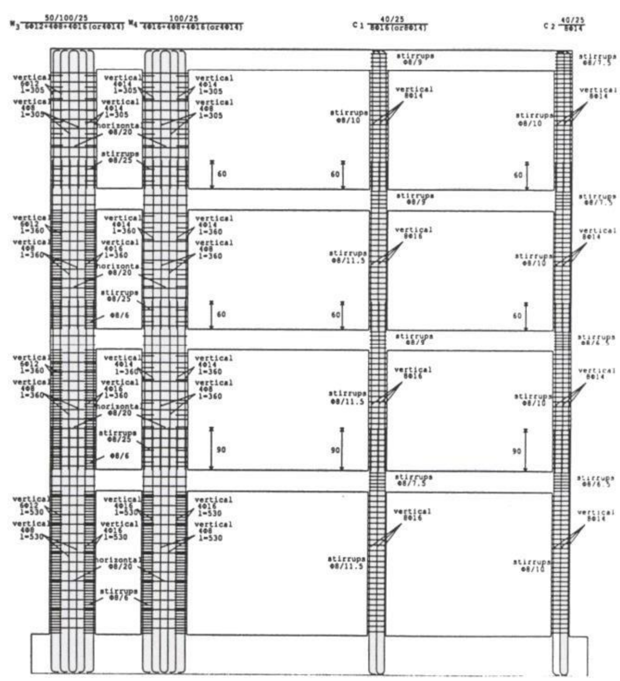


Figure 30: Reinforcement in Beams and Columns (EC-8)



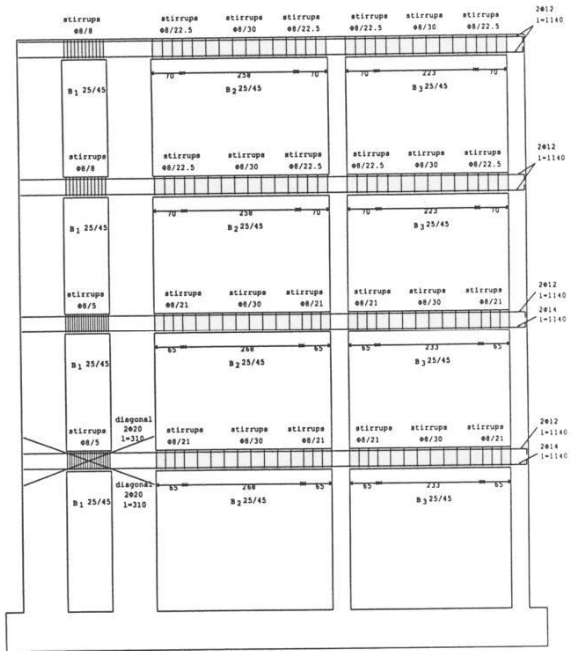


Figure 31:Reinforcement in Beams and Columns (DBD)

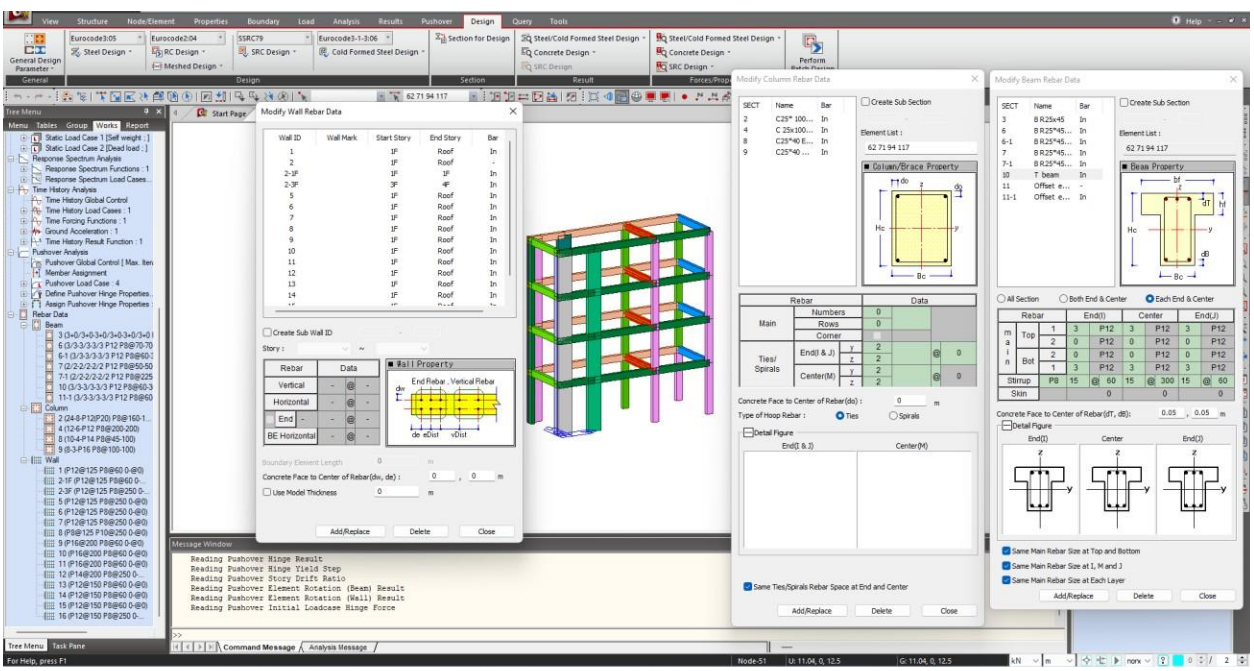


Figure 32: MIDAS Cross sectional Reinforcement detailing

After the definition of material and the type of plastic hinges, they were assigned to relevant members.

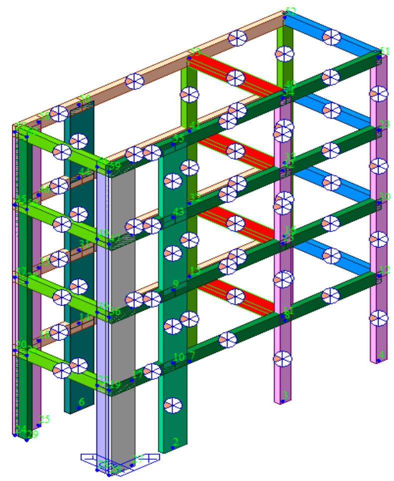


Figure 33: Plastic Hinges

4.4 Pushover Curve/Capacity Curve:

The modal and uniform loading curves were superimposed with the experimental capacity curve and is shown below.

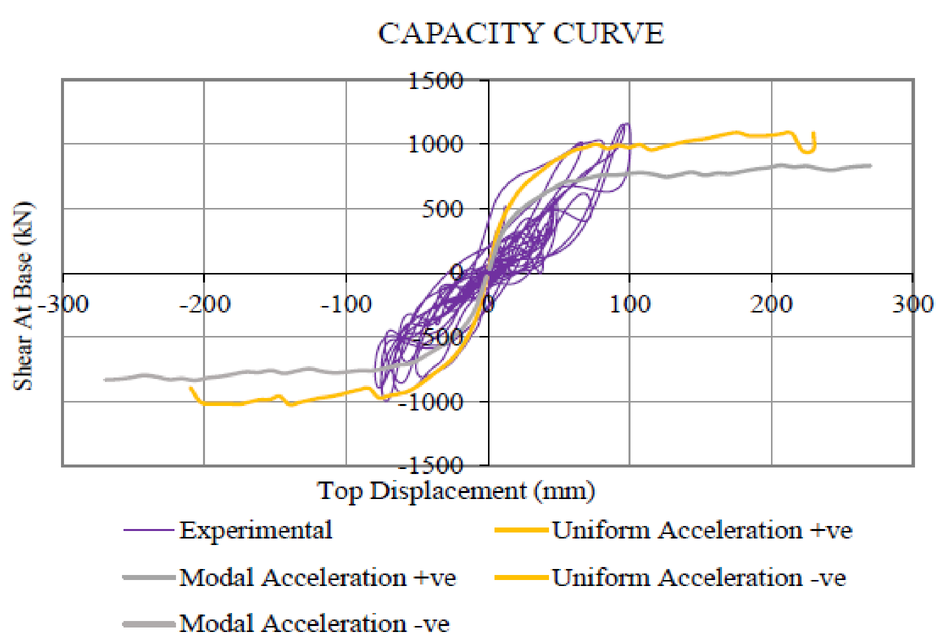


Figure 34: pushover capacity curves

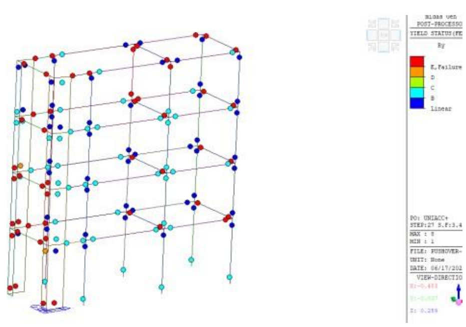
Table 27: Pushover Experimental vs Modelling Results comparison

	Shear Force (KN)		Displacement (mm)	
LOAD CASES	**Model**	**Experiment**	**Model**	**Experiment**
Uniform Acceleration +ve	1090.87	1161.9	230.00	100.689
Uniform Acceleration -ve	836.04	1161.9	-270.00	-79.76
Modal +ve	1025.45	1161.9	210.00	100.689
Modal -ve	836.04	1161.9	-270.00	-79.76

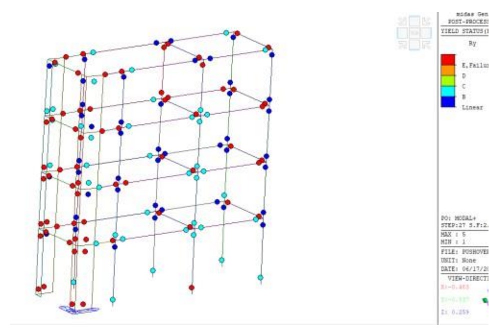
It can be observed that the values of shear forces especially in the positive direction remained very close the experimental ones, so the structure real behavior is adequately reproduced however the deviations in the displacement model remained quite large compared to the experimental ones which is an indication of low usage of the ductile resources of the structure during experiment.

4.5Plastic Hinge Status Results:

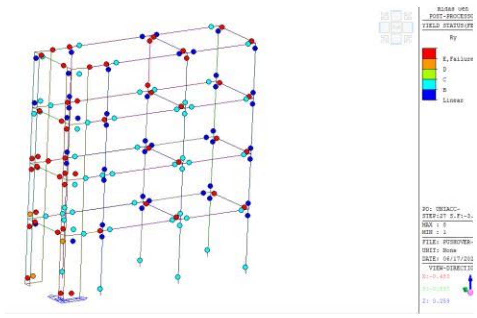
The pushover hinge yield status report for time step 27 for all load cases are presented below.



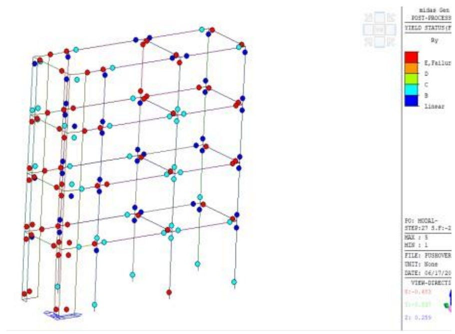
a. Uniform acceleration +ve



b. Modal Acc. +ve



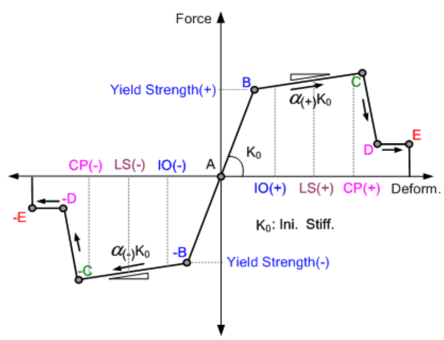
c. Uniform acceleration -ve



d. Modal Acc. +ve

Figure 35: Plastic Hinge status for Load Cases

Where the relative points of linear and up to failure on the hinge report corresponds to the following points on the FEMA curve.



It can be observed that in all cases the failure hinges appeared at the transverse beams or near the shear walls classifying them as the critical cross sections in our structure.

CONCLUSION:

The project comprised of modelling, an R.C. frame in MIDAS Gen, the analysis and comparison of the numerical model to an experimental model of the frame under high and low excitation.

The purpose was to cover the aspects of modelling a Reinforced Concrete structure and integration of the key parameters influencing the model.

The first part of the project comprised of representing the correct global behavior in terms of natural modal frequencies/Periods by comparing it with experimental values. For this purpose, the error was reduced to 2% by changing the model in different parameters from member's geometry to rigidity factors of members. A few observations include,

- Element positioning greatly influences the vibration frequencies. The software packages associate the element positioning on the grid based on their centroidal line associating no eccentricities near the nodes of different connecting members. The first part was greatly oriented towards the beams/panel offset to allow for the discrepancies/ eccentricities of practical structures.
- The monolithic casting of beams and slabs allows for the introduction of slab into the stiffness contribution for the structure but the effect on error reduction was minute practically allowing for the beams to be modelled completely as rectangular members as well.
- As opposed to the lumped mass approach, the software package converts directly the self-weights and applied loads into masses and inertial moments and associates them to a master node on each floor. However, if correctly modelled both the choices don't lead to any differences.
- Changing the nodal stiffness had the highest effect on increasing the stiffness of the structure and reducing the error in frequencies from 19% to almost 4%. The reduction is observed through reduced geometry (length of the members) near the nodes.

The main aspect of modelling as deduced from the project is stiffness. In the first part of the project lower nodal frequencies related to lower stiffness values and hence required increasing the global stiffness. On the contrary in the second part when response under high excitation was required and ductile behavior was to be introduced in the structure, stiffness (flexural and Shear) were reduced to account for the cracking effects.

Considering displacements, the error concerning the experimental data has been progressively reduced, reaching the final model value of about 2% of error. This demonstrates how the deformations induced by the earthquake lead the reinforced concrete sections to crack, resulting in high reductions in rigidity. Concerning the shear at the base, it was not possible to have an improvement of the results, reaching for the final model value of around 50% of error. Compare experimental values with the numerical ones, it resulted in the updated behavior factor equal to 2.35. So the error related to the base shear was about 3%.

At the end, a non-linear Pushover analysis was performed on the structure to obtain the capacity curves which were compared to the experimental base shear and max displacement values.

As for the last displacement, this turned out to be almost twice the experimental value, so it can be deduced that during the experiment only a part of the ductile resources of the structure was exploited.

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